Problem A. Abiyoyo

"Abiyoyo, Abiyoyo."

"Abiyoyo, Abiyoyo."

"Abiyoyo, yo yoyo yo yoyo."

Once upon a time there was a little boy, who played a ukulele. He'd go around town: clink, clunk, CLONK! Of course, the grownups would be busy, and they would say: "Take that thing out of here. We are talking. Git!" And they would kick him out of the house.

Now, in this town they used to tell stories. The old people used to tell stories about the monsters and giants that lived in the old days! They used to tell a story about Abiyoyo. They said he was as tall as a house, and could eat people up. Of course, nobody believed it, but they told the stories anyway.

But one day, the sun rose, blood red over the hill, and the first people that got up and looked out of their window. They saw a great big shadow in front of the sun, and they could feel the whole ground shake.

Women screamed.

Strong men fainted.

They said: "Run for your lives! Abiyoyo's coming!" Down through the fields he came. He came to the sheep, pasture and grabs a whole sheep. Yeowp! He eats it down in one bite. He comes to the cow pasture. Yuhk!

Just then the boy woke up. The boy rubbed his eyes and said: "Hey, what is coming over the fields? Oh, it is Abiyoyo." He grabbed his ukulele. Over the fields he went, right up to where Abiyoyo was. People screamed "Don't go near him! He will eat you alive!"

There was Abiyoyo. He had long fingernails because he never cut them. He had slobbery teeth because he never brushed them. Matted hair because he never combed it. Stinking feet because he never washed them. He was just about to come down with his claws, when the boy whipped out his ukulele.

Abiyoyo, Abiyoyo Abiyoyo, Abiyoyo Abiyoyo, yo yoyo yo yoyo Abiyoyo, yo yoyo yo yoyo.

Well, the monster had never heard a song about himself before, and a foolish grin spread across his face. And started to dance.

Abiyoyo, Abiyoyo.

The boy went faster.

Abiyoyo, yo yoyo, yo yoyo Abiyoyo, yoyo yoyo, ho ho ho ho ho ho ho ho.

The giant got out of breath. He staggered. He fell down flat on the ground.

"Booooom!" A hunter fired, and the Abiyoyo died. People streamed out of their houses, and ran across the fields. **They sang many times:**

Abiyoyo, Abiyoyo.

Then they all sang:

Abiyoyo, yo yoyo yo yoyo.

Abiyoyo, yo yoyo yo yoyo.

Input

This problem has several test cases. The first line of the input contains an integer t ($1 \le t \le 20$) which is the number of test cases.

Then t cases follow. For each test, a line contains an integer k ($1 \le k \le 20$) which means that people sang k times:

Abiyoyo, Abiyoyo.

Output

For each case, output all what they sang after the death of Abiyoyo in several lines.

Sample

3 2 5 1	Abiyoyo, Abiyoyo. Abiyoyo, Abiyoyo. Abiyoyo, yo yoyo yo yoyo. Abiyoyo, yo yoyo yo yoyo. Abiyoyo, Abiyoyo. Abiyoyo, Abiyoyo. Abiyoyo, Abiyoyo. Abiyoyo, Abiyoyo. Abiyoyo, Abiyoyo. Abiyoyo, yo yoyo yo yoyo. Abiyoyo, yo yoyo yo yoyo. Abiyoyo, Abiyoyo. Abiyoyo, yo yoyo yo yoyo. Abiyoyo, yo yoyo yo yoyo.

Problem B. Attacker-Defender Game

In the game of Attacker-Defender, the defender has h health points and d energy points at the beginning, and the attacker has a energy points.

In each round of the game, the attacker and the defender **move simultaneously**. In the attacker's (resp. defender's) move, he shows x (resp. y) points of his energy. If $x > y$, then the defender's health points decreases by 1 and the attacker's energy points decreases by x; if $x < y$, then the defender's health points increases by 1 and his energy points decreases by y; if $x = y$, then both players' energy points decrease by x or y.

When one of the following two events happens, the game ends.

- The defender has no health points: The attacker wins.
- The attacker's energy points is less than the defender's health points: The defender wins.

Both of them play with the strategies such that their winning probabilities are maximized when the opponent is playing optimally. Your task is to calculate their winning probabilities.

Input

The first line is the number of test cases which is less than 1200. For each test case, there is a line containing three integers h, d and a, which are all positive and less than 50.

Output

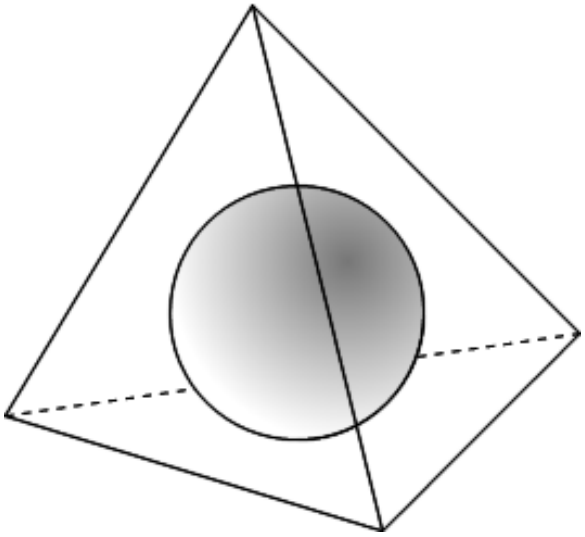
For each test case, output the attacker's winning probability with the precision of 6 digits.

Sample

3 1 1 1 1 2 2 1 3 3	0.000000 0.500000 0.750000

This page intentionally left blank.

Problem C. The Ball



In the three dimensional Euclidean space (X, Y, Z), the intersection of several half spaces and $\{X \geq 0, Y \geq 0, Z \geq 0\}$ forms an area **with positive volume**.

Here each half space is represented as a linear inequation $AX + BY + CZ \leq D$. Our problem is to find the largest available ball **fully locating in the area**.

Input

The input contains several test cases. The first line of input contains an integer T ($1 \leq T \leq 160$) indicating the number of cases.

For each case, the first line contains an integer N ($1 \leq N \leq 100$) indicating the number of half spaces. Each of the following lines describes a half space given by four integers A, B, C and D corresponding to the linear inequation $AX + BY + CZ \leq D$, where $-100 \leq A, B, C, D \leq 100$. The summation of N in input is up to 6200.

Output

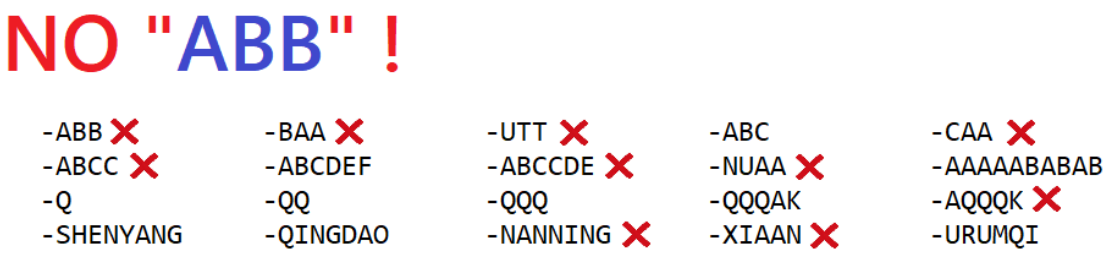
For each test case, output a line. If the size of available balls is unrestricted, output "Infinity". Else, output the largest radius of an available ball with the precision of 4 digits after the decimal point.

Sample

5 3 1 0 0 1 0 1 0 1 0 0 1 1 1 1 1 1 1 2 -1 -1 -1 -2 1 2 3 7 2 1 0 0 1 0 0 1 1 1 1 -1 0 0	0.5000 0.2113 0.5901 0.5000 Infinity

This page intentionally left blank.

Problem D. Banned Patterns



The new censorship law has been passed and a mass of characters and strings are explicitly prohibited appearing in the public publication.

Specifically, N patterns were blacklisted. To simplify the problem, we only consider the uppercase letters and each pattern is a string that consists of only uppercase letters.

The authority makes a total ban on those strings which have at least one substring matching a banned pattern.

If a substring can be transmuted to a pattern by choosing a permutation of all 26 uppercase letters, the matching satisfies the correctness. Here we list some examples.

Each two of ABB, ACC, BAA, UTT, XZZ and ZAA match each other.

ABCDECBA matches TYQAXQYT but does not match QWEWSEWQ.

You aim to design an efficient algorithm to judge the validity of upcoming publications.

Input

The first line contains a number T ($1 \leq T \leq 20$) indicating the number of test cases.

For each test case, the first line is the number of banned patterns N ($1 \leq N \leq 5000$).

Each of the following N lines contains a pattern in capital letters.

The $(N+2)$-th line contains an integer M ($1 \leq M \leq 250000$) which is the number of strings asked to be judged based on above patterns.

Each of the following M lines consists a string in capital letters as well.

The total length of patterns in a test case should not be larger than 100000, and the total length of strings asked to be judged should not be larger than 1000000.

Output

For each test case, output a line with the index and M letters. Then for each given string, **output "Y" if it is prohibited**, or **"N" if it is allowed**.

Sample

2 2 AAB ABB 8 TUU ZZY UVW ABABABA UUVV TTUUO EEEE XY 1 ABCBD 5 UVWXY UVVVY UVWVY YVWVY VVWVY	Case #1: Y Y N N Y Y N N Case #2: N N Y N N

Problem E. The Champion

Bob is attending a chess competition. Now the competition is in the knockout phase. There are 2^r players now, and they will play over r rounds.

In each knockout round, the remaining players would be divided into pairs, and the winner of each pair would advance to the next knockout round. Finally only one player would remain and be declared the champion.

Bob has already assigned all players in an order while he assigned himself to the **k-th site**. A better player is located at **a site with a smaller number** indicating a **higher order**. The probability that **a player with higher order wins a player with lower order** is p ($0 \le p \le 1$).

Bob notices that arrangement of matches is crucial for the final result.

For example, if there are 4 players and Bob is the second best player (he is the second site), and $p = 0.9$. In the first round, if Bob meets the best player, he will have only $0.1 \times 0.9 = 0.09$ probability to become the champion. However if he does not meet the best player in the first round, he will have $0.9 \times (0.9 \times 0.1 + 0.1 \times 0.9) = 0.162$ probability to become the champion. Now Bob wants to know, what is the winning probability for him in the best arrangement of matches.

Input

The first line in the input contains an integer t ($1 \le t \le 63000$) which is the number of test cases.

For each case, there is only one line containing two integers r and k ($1 \le r < 64, 1 \le k \le 2^r$) and a float-point number p ($0 \le p \le 1$) as described above.

Output

For each case, calculate the winning probability for Bob in the best arrangement. Output the probability with the precision of 6 digits.

Sample

2 1 1 0.8 2 2 0.9	0.800000 0.162000

This page intentionally left blank.

Problem F. The Chosen One

Welcome to the 2017 ACM-ICPC Asia Nanning Regional Contest.

Here is a breaking news. Now you have a chance to meet alone with the Asia Director through a game.

All boys and girls who chase their dreams have to stand in a line. They are given the numbers in the order in which they stand starting from 1.

The host then removes all boys and girls that are standing at an odd position with several rounds.

For example if there are $n = 8$ boys and girls in total. Initially standing people have numbers $1, 2, 3, 4, 5, 6, 7$ and 8. After the first round, people left are $2, 4, 6$ and 8. After the second round, only two people, 4 and 8, are still there.

The one who stays until the end is the chosen one.

I know you want to become the chosen one to meet alone with your idol. Given the number of boys and girls **in total**, can you find the best place to stand in the line so that you would become the chosen one?

Input

First line of the input contains the number of test cases t ($1 \leq t \leq 1000$).

Each of the next t lines contains the integer n which is the number of boys and girls in total, where $2 \leq n \leq 10^{50}$.

Output

The output displays t lines, each containing a single integer which is the place where you would stand to win the chance.

Sample

4 5 12 23 35	4 8 16 32

This page intentionally left blank.

Problem G. Five Dimensional Discrete Fourier Transform

The five dimensional discrete fourier transform over a $n_1 \times n_2 \times n_3 \times n_4 \times n_5$ complex array $a[i_1][i_2][i_3][i_4][i_5]$ where $0 \le i_1 \le n_1 - 1$, $0 \le i_2 \le n_2 - 1$, $0 \le i_3 \le n_3 - 1$, $0 \le i_4 \le n_4 - 1$ and $0 \le i_5 \le n_5 - 1$ is given by:

$$A[j_1][j_2][j_3][j_4][j_5] = \sum_{i_1=0}^{n_1-1} \cdots \sum_{i_5=0}^{n_5-1} a[i_1][i_2][i_3][i_4][i_5] e^{-2\pi\sqrt{-1}(i_1 j_1/n_1 + \cdots + i_5 j_5/n_5)}$$

for $0 \le j_1 \le n_1 - 1, \cdots, 0 \le j_5 \le n_5 - 1$.

Now comes your turn. For a given real coefficient α, suppose

$$a[i_1][i_2][i_3][i_4][i_5] = (i_1 \ xor \ i_2 \ xor \ i_3 \ xor \ i_4 \ xor \ i_5) e^{\sqrt{-1}(i_1 - i_2 + i_3 - i_4 + i_5)\alpha}.$$

Please calculate the value of

$$\frac{1}{(n_1 n_2 n_3 n_4 n_5)^{1.5}} \sum_{i_1=0}^{n_1-1} \cdots \sum_{i_5=0}^{n_5-1} |Re(A[i_1][i_2][i_3][i_4][i_5])|,$$

where Re means the real part of a complex number and $|\cdots|$ means the absolute value of real number.

Input

The first line of the input contains an integer T ($1 \le T \le 200$) indicating the total number of test cases. For each test case, a line contains five integers n_1, n_2, n_3, n_4, n_5 where $1 \le n_1, n_2, n_3, n_4, n_5 \le 10$, and a float number α ($-100000 \le \alpha \le 100000$) with at most 10 significance digits.

Output

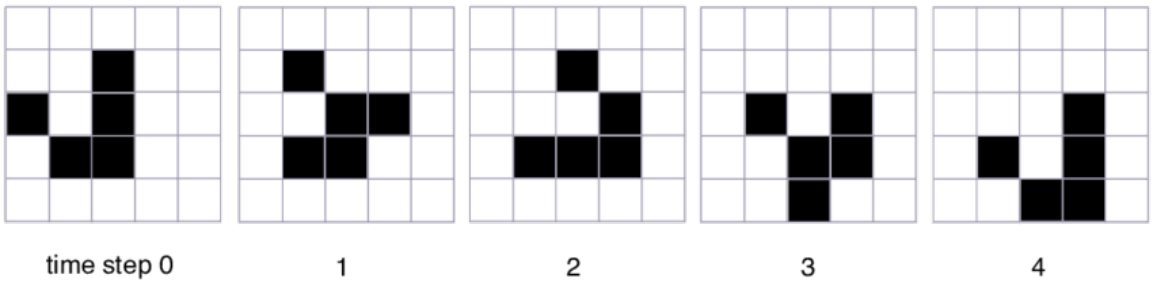
For each test case, output the value with the precision of 6 digits.

Sample

5 1 1 1 1 2 9.8 1 1 1 1 4 3.14 1 2 1 2 1 8.777 2 1 1 1 2 8.777 1 2 3 2 1 20.1314	0.657911 1.499989 0.398713 0.398713 0.942596

This page intentionally left blank.

Problem H. The Game of Life



From an evolutionary gene's-eye perspective, the genes are immortal, and our role, the meaning of life, is to perpetuate the genes. In a few centuries, all traces of our existence as human individuals, memories of us, all our accomplishments, will likely be gone and forgotten, except for genes that survive from those of us who successfully reproduced through the generations.

Of course, we do not experience the world from a gene's eye evolutionary perspective, but the Game of Life does. The universe of the Game of Life is an infinite two-dimensional orthogonal grid of square cells. Each cell is in one of two possible states, alive or dead. Every cell interacts with its eight neighbours, which are the cells that are horizontally, vertically or diagonally adjacent. At each step in time, the following transitions occur.

Any live cell with fewer than two live neighbours dies, as if caused by underpopulation.

Any live cell with two or three live neighbours lives on to the next generation.

Any live cell with more than three live neighbours dies, as if by overpopulation.

Any dead cell with exactly three live neighbours becomes a live cell, as if by reproduction.

The initial pattern, the 0-th generation, is given by a finite $N \times M$ grid where $N \leq 3$ and $M \leq 5$ as a part of the infinite grid. That implies that the state of any other grid is dead.

Genetic evolution is the meaning of biologic life, in that it is the why and how of it, as well as the stock of future biological existence. The trinity life in the game requests you to find answers of three problems.

At what point in the **first** 321 **generations, including the 0-th generation**, will the population be at its maximum?

And what is that maximum?

What will the population be at the 321-th generation?

Input

The first line of the input contains an integer t ($1 \leq t \leq 300$) specifying the number of test cases.

Each case consists of several lines. The first line contains two positive integers N and M. Each of the next N lines contains a string of length M, corresponding the initial pattern. We use "#" to describe a cell which is alive, and use "." to describe a cell which is dead.

Output

For each test case, output three integers a, b and c in a line, where a means the a-th generation, b is the population at that generation, and c is the population at the 321-th generation. **If there are several generations whose population meets the maximum, choose the smallest a for the output**.

Sample

3 3 3 ... ##. .#. 3 3 .#. ... ### 3 3 ... ### ...	1 4 4 10 20 12 0 3 3

Problem I. Rake It In

The designers have come up with a new simple game called "Rake It In". Two players, Alice and Bob, initially select an integer k and initialize a score indicator. An 4×4 board is created with 16 values placed on the board. **Starting with player Alice**, each player in a round selects a 2×2 region of the board, adding the sum of values in the region to the score indicator, and then rotating these four values 90 **degrees counterclockwise**.

After $2k$ rounds in total, each player has made decision in k times. The ultimate goal of **Alice** is to **maximize the final score**. However for **Bob**, his goal is to **minimize the final score**.

In order to test how good this game is, you are hired to write a program which can play the game. Specifically, given the starting configuration, they would like a program to determine the final score when **both players are entirely rational**.

Input

The input contains several test cases and the first line provides an integer t ($1 \leq t \leq 200$) which is the number of test cases.

Each case contains five lines. The first line provides the integer k ($1 \leq k \leq 3$). Each of the following four lines contains four integers indicating the values on the board initially. All values are integers between 1 to 10.

Output

For each case, output an integer in a line which is the predicted final score.

Sample

4 1 1 1 2 2 1 1 2 2 3 3 4 4 3 3 4 4 2 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 3 1 1 4 4 4 4 1 1 1 1 4 4 1 4 1 4 3 1 2 3 4 5 1 2 3 4 5 1 2 3 4 5 1	20 40 63 71

This page intentionally left blank.

Problem J. Rearrangement

In a two dimensional array of integers of size $2 \times n$, is it possible to rearrange integers so that the sum of two adjacent elements (which are adjacent in a common row or a common column) is never divisible by three?

Input

The input has several test cases and the first line contains an integer t ($1 \leq t \leq 200$) which is the number of test cases.

In each case, the first line contains an integers n ($1 \leq n \leq 10000$) indicating the number of columns in the array. The second line contains the elements of the array in the first row separated by single spaces. The third line contains the elements of the array in the second row separated by single spaces. The elements will be positive integers less then 1000000.

Output

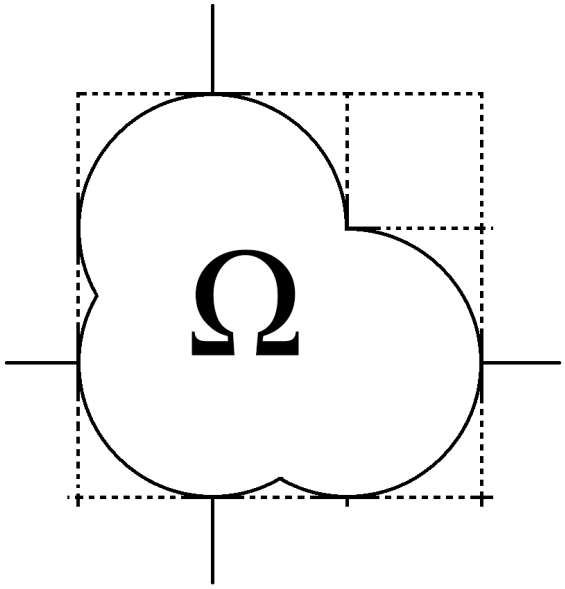
For each test case, output "YES" in a single line if any valid rearrangement exists, or "NO" if not.

Sample

6 3 3 6 9 1 4 7 3 3 6 9 1 3 8 5 1 2 3 4 5 6 7 8 9 10 10 1 1 1 1 1 1 1 1 1 1 2 3 2 3 2 3 2 3 2 3 2 3 1 2 3 2 3 1 1 2	YES NO YES YES YES NO

This page intentionally left blank.

Problem K. Resonators



The Sun-glade Isle as a portal is an important military fortress, located in the East China Sea. Several finest agents climbed onto the island successfully. They captured the portal and declared the victory of the battle, temporarily.

To reinforce the portal, they deployed N resonators around the portal. The centre of the portal is located at $(0, 0)$. A resonator occupies a circular area. The centre of the i-th one is (x_i, y_i) and its radius is r_i. These circular areas are likely to overlap.

The total defense capability is the sum of squared distances to the centre of the portal $(0, 0)$ from the locations which be occupied by at least one resonator. That is say that if there is a sufficient small region occupied by at least one resonator, this region would offer the defense capability up to the product of the area, and the squared distance to the centre $(0, 0)$.

If we suppose Ω as the union of all resonators which is the union of several circles. The total defense capability of this portal is actually the result of the following integral:

$$\int_{\Omega} |\mathbf{v}|^2 ds = \int_{\Omega} (x^2 + y^2) dx dy.$$

Here is your mission, as a super leader the Intel. Compute the total defense capability of this portal.

Input

The input contains several test cases and the first line of input gives the number of test cases which is up to 50.

For each test case, the first line contains the integer N ($N \leq 1000$) which is the total number of resonates deployed. The following N lines describe these resonators and the i-th line contains three integers x_i, y_i and r_i, where $|x_i|, |y_i| \leq 100$ and $1 \leq r_i \leq 5$.

Output

For each test case output the defense capability rounding up to three decimal places.

Sample

3 1 0 0 1 2 0 0 1 1 0 1 3 0 0 1 1 0 1 0 1 1	1.571 5.704 9.668

Hint

For the first test case in the sample, the answer if

$$\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}(x^2+y^2)dydx=\frac{\pi}{2}\approx 1.5708.$$

The answer of the second test case is the sum of

$$\int_{-1}^{1/2}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}(x^2+y^2)dydx\approx 1.15545$$

and

$$\int_{1/2}^{2}\int_{-\sqrt{1-(x-1)^2}}^{\sqrt{1-(x-1)^2}}(x^2+y^2)dydx\approx 4.54888.$$

Problem L. Twice Equation

For given L, find the smallest n no smaller than L for which there exists an positive integer m for which $2m(m+1) = n(n+1)$.

Input

This problem contains multiple test cases. The first line of a multiple input is an integer T ($1 \leq T < 1000$) followed by T input lines. Each line contains an integer L ($1 \leq L < 10^{190}$).

Output

For each given L, output the smallest n. If available n does not exist, output -1.

Sample

3 1 4 21	3 20 119

This page intentionally left blank.

Problem M. The Maximum Unreachable Node Set

In this problem, we would like to talk about unreachable sets of a **directed acyclic graph** $G = (V, E)$. In mathematics a directed acyclic graph (DAG) is a directed graph with no directed cycles. That is a graph such that there is no way to start at any node and follow a consistently-directed sequence of edges in E that eventually loops back to the beginning again.

A node set denoted by $V_{UR} \subset V$ containing several nodes is known as an unreachable node set of G if, for each two different nodes u and v in V_{UR}, there is no way to start at u and follow a consistently-directed sequence of edges in E that finally archives the node v. You are asked in this problem to calculate the size of the **maximum unreachable node set** of a given graph G.

Input

The input contains several test cases and the first line contains an integer T ($1 \leq T \leq 500$) which is the number of test cases.

For each case, the first line contains two integers n ($1 \leq n \leq 100$) and m ($0 \leq m \leq n(n-1)/2$) indicating the number of nodes and the number of edges in the graph G. Each of the following m lines describes a directed edge with two integers u and v ($1 \leq u, v \leq n$ *and* $u \neq v$) indicating an edge from the u-th node to the v-th node. All edges provided in this case are distinct.

We guarantee that all directed graphs given in input are DAGs and the sum of m in input is smaller than 500000.

Output

For each test case, output an integer in a line which is the size of the maximum unreachable node set of G.

Sample

3 4 4 1 2 1 3 2 4 3 4 4 3 1 2 2 3 3 4 6 5 1 2 4 2 6 2 2 3 2 5	2 1 3

This page intentionally left blank.